

Anime

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2025-11-22

Construction des 2 tableaux

Nous avons choisi de traiter un par un les fichiers csv pour ensuite les concaténer en 2 tableaux afin de répondre aux problématiques.

Fonctions utiles pour la suite

Voici des fonctions que nous avons ensuite utilisé pour traiter plus simplement les csv.

```
suppressPackageStartupMessages({
  library(data.table)
  library(dplyr)
  library(tidyr)
  library(stringr)
  library(lubridate)
  library(purrr)
})

# parse_list_col : nettoie une colonne contenant des listes sous forme de
# chaînes de caractères (ex"['a','b']").
# et renvoie une liste de vecteurs propres.
parse_list_col <- function(x) {
  x <- as.character(x)
  x[is.na(x)] <- "[]"
  x <- str_replace_all(x, "\\[[|\\]|\\'", "")
  x <- str_trim(x)
  out <- str_split(x, ",")
  out <- lapply(out, function(v) {
    v <- str_trim(v)
    v[v != ""]
  })
  out
}

# multihot_topk :
# - df : data frame
# - id_col : nom de la colonne identifiant
# - list_col : colonne contenant des listes
# - prefix : préfixe des nouvelles colonnes
# - top_k : garde seulement les k éléments les plus fréquents
# Produit une matrice multihot.
```

```

multihot_topk <- function(df, id_col, list_col, prefix, top_k = NULL) {
  lst <- parse_list_col(df[[list_col]])
  all_levels <- unlist(lst)
  all_levels <- all_levels[all_levels != ""]
  if (length(all_levels) == 0) return(df[, ..id_col, drop = FALSE])

  freq <- sort(table(all_levels), decreasing = TRUE)
  levels_keep <- names(freq)
  if (!is.null(top_k)) levels_keep <- levels_keep[1:min(top_k, length(levels_keep))]

  bin_mat <- sapply(levels_keep, function(level) {
    as.integer(map_lgl(lst, ~ level %in% .x))
  })
  bin_df <- as.data.frame(bin_mat)
  names(bin_df) <- paste0(prefix, "_", make.names(levels_keep))
  bind_cols(df[, ..id_col, drop = FALSE], bin_df)
}

# onehot_topk :
# - col : colonne catégorielle simple
# Transforme en one-hot encoding, réduit aux catégories les plus fréquentes.
onehot_topk <- function(df, col, prefix, top_k = NULL) {
  v <- as.character(df[[col]])
  if (!is.null(top_k)) {
    tab <- sort(table(v), decreasing = TRUE)
    keep <- names(tab)[seq_len(min(top_k, length(tab)))]
    v2 <- ifelse(v %in% keep, v, "Other")
  } else {
    # top_k = NULL : on garde toutes les modalités originales
    v2 <- v
  }
  mm <- model.matrix(~ v2 - 1)
  mm <- as.data.frame(mm)
  names(mm) <- paste0(prefix, "_", sub("^v2", "", names(mm)))
  dplyr::bind_cols(df %>% dplyr::select(-all_of(col)), mm)
}

# to_num : conversion numérique silencieuse (NA si impossible)
to_num <- function(x) suppressWarnings(as.numeric(x))

```

On traite ensuite 1 par 1 les fichiers csv :

character__anime__works.csv

```

char_works <- fread("character_anime_works.csv")
# Remplace les points dans les noms de colonnes par des underscores
names(char_works) <- str_replace_all(names(char_works), "\\.", "_")

char_features <- char_works %>%
  group_by(anime_mal_id) %>%

```

```
summarise(
  num_characters = n_distinct(character_mal_id),
  num_main_characters = n_distinct(character_mal_id[role == "Main"]),
  num_supporting_characters = n_distinct(character_mal_id[role == "Supporting"]),
  .groups = "drop"
)
```

```
head(char_works)
```

```
##   anime_mal_id character_mal_id character_name    role
##   <int>          <int>          <char>    <char>
## 1:      2928        5781        Atoli      Main
## 2:      2928         33        Haseo      Main
## 3:      2928         32         Ovan      Main
## 4:      2928         34        Shino      Main
## 5:      2928        5785         Aina    Supporting
## 6:      2928        5784    Endrance    Supporting
```

```
head(char_features)
```

```
## # A tibble: 6 x 4
##   anime_mal_id num_characters num_main_characters num_supporting_characters
##   <int>          <int>          <int>          <int>
## 1         1         126             4             122
## 2         5          42             6             36
## 3         6          97             4             93
## 4         7          53             6             47
## 5         8          10             5              5
## 6        15          64             5             59
```

Avec le tableau `char_features`, on obtient pour un anime le nombre de personnages, ainsi que le nombre de personnages principaux et secondaires.

character_nicknames.csv

```
char_nicks <- fread("character_nicknames.csv")
names(char_nicks) <- str_replace_all(names(char_nicks), "\\.", "_")

char_nick_feat <- char_nicks %>%
  mutate(character_mal_id = as.character(character_mal_id)) %>%
  group_by(character_mal_id) %>%
  summarise(num_nicknames = n(), .groups = "drop")
head(char_nick_feat)
```

```
## # A tibble: 6 x 2
##   character_mal_id num_nicknames
##   <chr>          <int>
## 1 100023             4
## 2 100053             1
## 3 10006             1
## 4 10007             1
## 5 100073            1
## 6 10010             2
```

Pour personnage, on obtient son nombre de nicknames.

person_altername_names.csv

```
person_alt_names <- fread("person_altername_names.csv")
names(person_alt_names) <- str_replace_all(names(person_alt_names), "\\.", "_")
```

On n'a pas utilisé ce csv par la suite.

recommendations.csv

```
reco <- fread("recommendations.csv")
names(reco) <- str_replace_all(names(reco), "\\.", "_")

reco_features <- reco %>%
  group_by(mal_id) %>%
  summarise(
    num_recommendations_made = n(),
    .groups = "drop"
  )

reco_features <- reco_features %>% rename(anime_mal_id = mal_id)

head(reco_features)
```

```
## # A tibble: 6 x 2
##   anime_mal_id num_recommendations_made
##         <int>             <int>
## 1             1             165
## 2             5              10
## 3             6            120
## 4             7             42
## 5             8              10
## 6            15             32
```

On obtient pour chaque anime le nombre de recommandations faites par cet anime.

stats.csv

```
stats <- fread("stats.csv")
names(stats) <- str_replace_all(names(stats), "\\.", "_")

stats <- stats %>% rename(anime_mal_id = mal_id)

stats_features_anime <- stats %>%
  select(
    anime_mal_id,
    watching,
    completed,
    on_hold,
    dropped,
    plan_to_watch,
    total
  )

head(stats_features_anime)
```

	anime_mal_id	watching	completed	on_hold	dropped	plan_to_watch	total
	<int>	<int>	<int>	<int>	<int>	<int>	<int>
## 1:	59356	7	146	4	20	20	197
## 2:	56036	21	770	8	29	113	941
## 3:	2928	451	14953	302	349	6472	22527
## 4:	3269	726	22790	452	537	9762	34267
## 5:	4469	241	6918	182	266	3528	11135
## 6:	454	325	15421	261	423	5027	21457

On a choisi de garder pour chaque anime les colonnes suivantes : - watching, le nombre d'utilisateurs actuellement en train de regarder - completed, le nombre d'utilisateurs ayant terminé - on_hold, le nombre d'utilisateurs ayant mis en pause - dropped, le nombre d'utilisateurs ayant abandonné - plan_to_watch, le nombre d'utilisateurs prévoyant de regarder - total, total global

person_voice_works.csv

```
p_voice <- fread("person_voice_works.csv")
names(p_voice) <- str_replace_all(names(p_voice), "\\.", "_")

p_voice[, person_mal_id := as.character(person_mal_id)]
p_voice[, character_mal_id := as.character(character_mal_id)]
p_voice[, anime_mal_id := as.integer(anime_mal_id)]

# Supprime les doublons
p_voice <- p_voice %>% distinct()

voice_features <- p_voice %>%
  group_by(anime_mal_id) %>%
  summarise(
    num_doubleurs = n_distinct(person_mal_id),
    num_main_doubleurs = n_distinct(person_mal_id[role == "Main"]),
    num_support_doubleurs = n_distinct(person_mal_id[role == "Supporting"]),
    .groups = "drop"
  )
head(voice_features)
```

	anime_mal_id	num_doubleurs	num_main_doubleurs	num_support_doubleurs
	<int>	<int>	<int>	<int>
## 1	1	418	44	377
## 2	5	160	45	116
## 3	6	302	42	264
## 4	7	108	29	81
## 5	8	18	10	9
## 6	15	189	31	163

On a pour chaque anime le nombre de doubleurs ainsi que le nombre de doubleurs de personnages principaux et secondaires.

profiles.csv

```
profiles <- fread("profiles.csv")
names(profiles) <- str_replace_all(names(profiles), "\\.", "_")
```

```

profiles <- profiles %>%
  mutate(
    gender = tolower(gender),
    gender = ifelse(gender %in% c("male","female","non-binary"), gender, "unknown")
  )

birth_year <- str_extract(profiles$birthday, "[0-9]{4}")
birth_year_num <- to_num(birth_year)

profiles <- profiles %>%
  mutate(
    birth_year = birth_year_num,
    age = ifelse(!is.na(birth_year), year(Sys.Date()) - birth_year, NA_real_),
    age = ifelse(age >= 8 & age <= 100, age, NA_real_)
  )

profiles_user <- profiles %>%
  transmute(
    username,
    age,
    gender,
    prof_on_hold = to_num(on_hold),
    prof_dropped = to_num(dropped),
    prof_plan_to_watch = to_num(plan_to_watch)
  ) %>%
  onehot_topk("gender", "gender", top_k = NULL)
head(profiles_user)

```

```

##      username  age prof_on_hold prof_dropped prof_plan_to_watch gender_female
##      <char> <num>      <num>      <num>      <num>      <num>
## 1:  ishikawas   NA         NA         NA         NA         0
## 2:    CKK2     NA         15         0        405         0
## 3:  -----788  NA         0         0         1         1
## 4:  potatoaris  NA         0         0         4         0
## 5:   Rinrintan  NA        40        16        34         0
## 6:   Karincakes NA        22        17       508         0
##      gender_male gender_non-binary gender_unknown
##      <num>      <num>      <num>
## 1:         0         0         1
## 2:         0         0         1
## 3:         0         0         0
## 4:         0         0         1
## 5:         0         0         1
## 6:         0         0         1

```

On obtient un tableau des utilisateurs avec une colonne pour l'âge (estimé à partir de l'année de naissance), plusieurs indicateurs sur le comportement de l'utilisateur (nombre d'animes mis en pause, abandonnés ou prévus), ainsi que le genre. Ce dernier est traité par un encodage one-hot, c'est-à-dire qu'une colonne binaire est créée pour chaque valeur possible du genre (male, female, non_binary, unknown). Pour chaque utilisateur, la colonne correspondant à son genre vaut 1 et les autres valent 0.

person_anime_works.csv

```
p_anime <- fread("person_anime_works.csv")
names(p_anime) <- str_replace_all(names(p_anime), "\\.", "_")
p_anime <- p_anime %>%
  mutate(
    person_mal_id = as.character(person_mal_id),
    anime_mal_id = as.integer(anime_mal_id)
  )

# Extraction des features liées au staff musical par anime
person_anime_features <- p_anime %>%
  mutate(
    position_clean = str_to_lower(position),
    is_op = str_detect(position_clean, "\\bop\\b|opening|theme song performance \\(op\\)|tv op"),
    is_ed = str_detect(position_clean, "\\bed\\b|ending|theme song performance \\(ed\\)",
    is_insert = str_detect(position_clean, "inserted song")
  ) %>%
  group_by(anime_mal_id) %>%
  summarise(
    num_music_people = n_distinct(person_mal_id),
    num_music_entries = n(),
    num_op_entries = sum(is_op, na.rm = TRUE),
    num_ed_entries = sum(is_ed, na.rm = TRUE),
    num_insert_entries = sum(is_insert, na.rm = TRUE),
    .groups = "drop"
  )
head(person_anime_features)
```

```
## # A tibble: 6 x 6
##   anime_mal_id num_music_people num_music_entries num_op_entries num_ed_entries
##         <int>         <int>         <int>         <int>         <int>
## 1             1             120             120              7              1
## 2             5             124             124              5              2
## 3             6              69              69              2              0
## 4             7              66              66             10              0
## 5             8              30              30              0              0
## 6            15             266             266              3              0
## # i 1 more variable: num_insert_entries <int>
```

details.csv

```
details <- fread("details.csv")
names(details) <- str_replace_all(names(details), "\\.", "_")

details <- details %>% rename(anime_mal_id = mal_id)

details_base <- details %>%
  transmute(
    anime_mal_id,
    type, status, source, rating, season, year, episodes,
    score = to_num(score),
    scored_by = to_num(scored_by),
```

```

rank = to_num(rank),
members = to_num(members),
favorites = to_num(favorites),
genres, studios, themes, demographics, producers,
explicit_genres, licensors, streaming
)

head(details_base)

##      anime_mal_id      type      status      source      rating
##      <int>      <char>      <char>      <char>      <char>
## 1:      59356      Movie Finished Airing Original      G - All Ages
## 2:      56036      Music Finished Airing Original PG-13 - Teens 13 or older
## 3:       2928       OVA Finished Airing      Game PG-13 - Teens 13 or older
## 4:       3269      Movie Finished Airing      Game PG-13 - Teens 13 or older
## 5:      4469 Special Finished Airing      Game PG-13 - Teens 13 or older
## 6:       454       OVA Finished Airing Original      R+ - Mild Nudity
##      season      year      episodes      score      scored_by      rank      members      favorites
##      <char>      <num>      <num>      <num>      <num>      <num>      <num>      <num>
## 1:      NA      1      NA      NA      17086      195      0
## 2:      NA      1      6.53      503      NA      941      2
## 3:      NA      1      6.65      9745      6366      22525      31
## 4:      NA      1      7.06      15373      4194      34264      104
## 5:      NA      1      6.35      4317      8182      11135      10
## 6:      NA      1      6.09      10021      9692      21458      20
##
##      genres      studios      themes
##      <char>      <char>      <char>
## 1:      ['Comedy']      []      []
## 2:      ['Horror', 'Supernatural']      ['Flat Studio']      ['Music']
## 3:      ['Adventure', 'Drama', 'Fantasy']      ['Bee Train']      ['Video Game']
## 4:      ['Action', 'Fantasy']      ['CyberConnect2']      ['Video Game']
## 5:      ['Comedy', 'Fantasy', 'Sci-Fi']      []      ['Parody', 'Video Game']
## 6:      ['Comedy', 'Fantasy']      ['Bee Train']      ['Video Game']
##      demographics      producers      explicit_genres
##      <char>      <char>      <char>
## 1:      []      ['Nagoya Zokei University']      []
## 2:      []      []      []
## 3:      []      ['Bandai Visual', 'CyberConnect2']      []
## 4:      []      ['Bandai Visual']      []
## 5:      []      ['Bandai Visual']      []
## 6:      []      ['CyberConnect2']      []
##      licensors      streaming
##      <char>      <char>
## 1:      []      []
## 2:      []      []
## 3:      []      []
## 4:      ['Funimation', 'Bandai Entertainment']      []
## 5:      []      []
## 6:      ['Bandai Entertainment']      []

details_numeric <- details_base %>%
  mutate(
    episodes = to_num(episodes),
    year = to_num(year),

```

```

) %>%
select(anime_mal_id, episodes, year)

head(details_numeric)

##      anime_mal_id episodes   year
##      <int>      <num> <num>
## 1:      59356         1    NA
## 2:      56036         1    NA
## 3:       2928         1    NA
## 4:       3269         1    NA
## 5:       4469         1    NA
## 6:        454         1    NA

details_cat <- details_base %>%
  select(anime_mal_id, type, status, source, rating, season) %>%
  onehot_topk("type", "type", top_k = 10) %>%
  onehot_topk("status", "status", top_k = NULL) %>%
  onehot_topk("source", "source", top_k = 15) %>%
  onehot_topk("rating", "rating", top_k = NULL) %>%
  onehot_topk("season", "season", top_k = NULL)

head(details_cat)

##      anime_mal_id type_type_CM type_Movie type_Music type_ONA type_OVA type_PV
##      <int> <num> <num> <num> <num> <num> <num>
## 1:      59356      0      0      1      0      0      0
## 2:      56036      0      0      0      1      0      0
## 3:       2928      0      0      0      0      1      0
## 4:       3269      0      0      1      0      0      0
## 5:       4469      0      0      0      0      0      0
## 6:        454      0      0      0      0      1      0
##      type_Special type_TV type_TV Special status_Currently Airing
##      <num> <num> <num> <num> <num>
## 1:      0      0      0      0      0
## 2:      0      0      0      0      0
## 3:      0      0      0      0      0
## 4:      0      0      0      0      0
## 5:      1      0      0      0      0
## 6:      0      0      0      0      0
##      status_Finished Airing status_Not yet aired source_4-koma manga source_Book
##      <num> <num> <num> <num> <num>
## 1:      1      0      0      0      0
## 2:      1      0      0      0      0
## 3:      1      0      0      0      0
## 4:      1      0      0      0      0
## 5:      1      0      0      0      0
## 6:      1      0      0      0      0
##      source_Game source_Light novel source_Manga source_Mixed media source_Music
##      <num> <num> <num> <num> <num>
## 1:      0      0      0      0      0
## 2:      0      0      0      0      0
## 3:      1      0      0      0      0
## 4:      1      0      0      0      0

```

```

## 5:          1          0          0          0          0
## 6:          0          0          0          0          0
##      source_Novel source_Original source_Other source_Picture book source_Unknown
##          <num>          <num>          <num>          <num>          <num>
## 1:          0          1          0          0          0
## 2:          0          1          0          0          0
## 3:          0          0          0          0          0
## 4:          0          0          0          0          0
## 5:          0          0          0          0          0
## 6:          0          1          0          0          0
##      source_Visual novel source_Web manga source_Web novel rating_
##          <num>          <num>          <num>          <num>
## 1:          0          0          0          0
## 2:          0          0          0          0
## 3:          0          0          0          0
## 4:          0          0          0          0
## 5:          0          0          0          0
## 6:          0          0          0          0
##      rating_G - All Ages rating_PG - Children rating_PG-13 - Teens 13 or older
##          <num>          <num>          <num>
## 1:          1          0          0
## 2:          0          0          1
## 3:          0          0          1
## 4:          0          0          1
## 5:          0          0          1
## 6:          0          0          0
##      rating_R - 17+ (violence & profanity) rating_R+ - Mild Nudity
##          <num>          <num>
## 1:          0          0
## 2:          0          0
## 3:          0          0
## 4:          0          0
## 5:          0          0
## 6:          0          1
##      rating_Rx - Hentai season_ season_fall season_spring season_summer
##          <num> <num>          <num>          <num>          <num>
## 1:          0      1          0          0          0
## 2:          0      1          0          0          0
## 3:          0      1          0          0          0
## 4:          0      1          0          0          0
## 5:          0      1          0          0          0
## 6:          0      1          0          0          0
##      season_winter
##          <num>
## 1:          0
## 2:          0
## 3:          0
## 4:          0
## 5:          0
## 6:          0
genres_mh <- multihot_topk(details_base, "anime_mal_id", "genres", "genre", top_k = 30)
themes_mh <- multihot_topk(details_base, "anime_mal_id", "themes", "theme", top_k = 25)
demo_mh   <- multihot_topk(details_base, "anime_mal_id", "demographics", "demo", top_k = NULL)

```

```
stud_mh <- multihot_topk(details_base, "anime_mal_id", "studios", "studio", top_k = 30)
prod_mh <- multihot_topk(details_base, "anime_mal_id", "producers", "producer", top_k = 30)
lic_mh <- multihot_topk(details_base, "anime_mal_id", "licensors", "licensor", top_k = 20)
stream_mh <- multihot_topk(details_base, "anime_mal_id", "streaming", "stream", top_k = 20)
```

```
details_features <- details_numeric %>%
  left_join(details_cat, by = "anime_mal_id") %>%
  left_join(genres_mh, by = "anime_mal_id") %>%
  left_join(themes_mh, by = "anime_mal_id") %>%
  left_join(demo_mh, by = "anime_mal_id") %>%
  left_join(stud_mh, by = "anime_mal_id") %>%
  left_join(prod_mh, by = "anime_mal_id") %>%
  left_join(lic_mh, by = "anime_mal_id") %>%
  left_join(stream_mh, by = "anime_mal_id")
```

```
head(details_features)
```

```
##      anime_mal_id episodes   year type_ type_CM type_Movie type_Music type_OVA
##      <int>      <num> <num> <num>      <num>      <num>      <num>      <num>
## 1:      59356         1    NA     0         0         1         0         0
## 2:      56036         1    NA     0         0         0         1         0
## 3:       2928         1    NA     0         0         0         0         0
## 4:       3269         1    NA     0         0         1         0         0
## 5:       4469         1    NA     0         0         0         0         0
## 6:        454         1    NA     0         0         0         0         0
##      type_OVA type_PV type_Special type_TV type_TV Special
##      <num>      <num>      <num>      <num>      <num>
## 1:         0         0         0         0         0
## 2:         0         0         0         0         0
## 3:         1         0         0         0         0
## 4:         0         0         0         0         0
## 5:         0         0         1         0         0
## 6:         1         0         0         0         0
##      status_Currently Airing status_Finished Airing status_Not yet aired
##      <num>      <num>      <num>      <num>      <num>
## 1:         0         1         0         1         0
## 2:         0         1         0         1         0
## 3:         0         1         0         1         0
## 4:         0         1         0         1         0
## 5:         0         1         0         1         0
## 6:         0         1         0         1         0
##      source_4-koma manga source_Book source_Game source_Light novel source_Manga
##      <num>      <num>      <num>      <num>      <num>      <num>
## 1:         0         0         0         0         0         0
## 2:         0         0         0         0         0         0
## 3:         0         0         1         0         0         0
## 4:         0         0         1         0         0         0
## 5:         0         0         1         0         0         0
## 6:         0         0         0         0         0         0
##      source_Mixed media source_Music source_Novel source_Original source_Other
##      <num>      <num>      <num>      <num>      <num>
## 1:         0         0         0         1         0
## 2:         0         0         0         1         0
## 3:         0         0         0         0         0
```

## 4:	0	0	0	0	0
## 5:	0	0	0	0	0
## 6:	0	0	0	1	0
##	source_Picture	book	source_Unknown	source_Visual	novel
##				novel	source_Web
##					manga
##		<num>	<num>	<num>	<num>
## 1:	0	0	0	0	0
## 2:	0	0	0	0	0
## 3:	0	0	0	0	0
## 4:	0	0	0	0	0
## 5:	0	0	0	0	0
## 6:	0	0	0	0	0
##	source_Web	novel	rating_	rating_G - All Ages	rating_PG - Children
##		<num>	<num>	<num>	<num>
## 1:	0	0	1	0	0
## 2:	0	0	0	0	0
## 3:	0	0	0	0	0
## 4:	0	0	0	0	0
## 5:	0	0	0	0	0
## 6:	0	0	0	0	0
##	rating_PG-13 - Teens 13 or older	rating_R - 17+ (violence & profanity)			
##		<num>		<num>	
## 1:		0		0	
## 2:		1		0	
## 3:		1		0	
## 4:		1		0	
## 5:		1		0	
## 6:		0		0	
##	rating_R+ - Mild Nudity	rating_Rx - Hentai	season_	season_fall	season_spring
##		<num>	<num>	<num>	<num>
## 1:	0	0	1	0	0
## 2:	0	0	1	0	0
## 3:	0	0	1	0	0
## 4:	0	0	1	0	0
## 5:	0	0	1	0	0
## 6:	1	0	1	0	0
##	season_summer	season_winter	genre_Comedy	genre_Fantasy	genre_Action
##		<num>	<num>	<int>	<int>
## 1:	0	0	1	0	0
## 2:	0	0	0	0	0
## 3:	0	0	0	1	0
## 4:	0	0	0	1	1
## 5:	0	0	1	1	0
## 6:	0	0	1	1	0
##	genre_Adventure	genre_Sci.Fi	genre_Drama	genre_Romance	genre_Slice.of.Life
##		<int>	<int>	<int>	<int>
## 1:	0	0	0	0	0
## 2:	0	0	0	0	0
## 3:	1	0	1	0	0
## 4:	0	0	0	0	0
## 5:	0	1	0	0	0
## 6:	0	0	0	0	0
##	genre_Hentai	genre_Supernatural	genre_Avant.Garde	genre_Mystery	genre_Ecchi
##		<int>	<int>	<int>	<int>
## 1:	0	0	0	0	0

```

## 2:      0      1      0      0      0
## 3:      0      0      0      0      0
## 4:      0      0      0      0      0
## 5:      0      0      0      0      0
## 6:      0      0      0      0      0
##  genre_Sports genre_Horror genre_Suspense genre_Award.Winning genre_Gourmet
##      <int>      <int>      <int>      <int>      <int>
## 1:      0      0      0      0      0
## 2:      0      1      0      0      0
## 3:      0      0      0      0      0
## 4:      0      0      0      0      0
## 5:      0      0      0      0      0
## 6:      0      0      0      0      0
##  genre_Boys.Love genre_Girls.Love genre_Erotica theme_Music theme_School
##      <int>      <int>      <int>      <int>      <int>
## 1:      0      0      0      0      0
## 2:      0      0      0      1      0
## 3:      0      0      0      0      0
## 4:      0      0      0      0      0
## 5:      0      0      0      0      0
## 6:      0      0      0      0      0
##  theme_Historical theme_Mecha theme_Anthropomorphic theme_Parody
##      <int>      <int>      <int>      <int>
## 1:      0      0      0      0
## 2:      0      0      0      0
## 3:      0      0      0      0
## 4:      0      0      0      0
## 5:      0      0      0      1
## 6:      0      0      0      0
##  theme_Adult.Cast theme_Military theme_Super.Power theme_Martial.Arts
##      <int>      <int>      <int>      <int>
## 1:      0      0      0      0
## 2:      0      0      0      0
## 3:      0      0      0      0
## 4:      0      0      0      0
## 5:      0      0      0      0
## 6:      0      0      0      0
##  theme_Space theme_Mythology theme_Harem theme_Isekai theme_Psychological
##      <int>      <int>      <int>      <int>      <int>
## 1:      0      0      0      0      0
## 2:      0      0      0      0      0
## 3:      0      0      0      0      0
## 4:      0      0      0      0      0
## 5:      0      0      0      0      0
## 6:      0      0      0      0      0
##  theme_Idols..Female. theme_Mahou.Shoujo theme_Strategy.Game
##      <int>      <int>      <int>
## 1:      0      0      0
## 2:      0      0      0
## 3:      0      0      0
## 4:      0      0      0
## 5:      0      0      0
## 6:      0      0      0
##  theme_Educational theme_Team.Sports theme_Detective theme_Gag.Humor

```

```

##          <int>          <int>          <int>          <int>
## 1:          0          0          0          0
## 2:          0          0          0          0
## 3:          0          0          0          0
## 4:          0          0          0          0
## 5:          0          0          0          0
## 6:          0          0          0          0
##  theme_Racing theme_Gore theme_CGDCT demo_Kids demo_Shounen demo_Seinen
##          <int>          <int>          <int>          <int>          <int>          <int>
## 1:          0          0          0          0          0          0
## 2:          0          0          0          0          0          0
## 3:          0          0          0          0          0          0
## 4:          0          0          0          0          0          0
## 5:          0          0          0          0          0          0
## 6:          0          0          0          0          0          0
##  demo_Shoujo demo_Josei studio_Toei.Animation studio_Sunrise studio_J.C.Staff
##          <int>          <int>          <int>          <int>          <int>
## 1:          0          0          0          0          0
## 2:          0          0          0          0          0
## 3:          0          0          0          0          0
## 4:          0          0          0          0          0
## 5:          0          0          0          0          0
## 6:          0          0          0          0          0
##  studio_TMS.Entertainment studio_Madhouse studio_Production.I.G
##          <int>          <int>          <int>
## 1:          0          0          0
## 2:          0          0          0
## 3:          0          0          0
## 4:          0          0          0
## 5:          0          0          0
## 6:          0          0          0
##  studio_Shanghai.Animation.Film.Studio studio_Studio.Deen studio_OLM
##          <int>          <int>          <int>
## 1:          0          0          0
## 2:          0          0          0
## 3:          0          0          0
## 4:          0          0          0
## 5:          0          0          0
## 6:          0          0          0
##  studio_Studio.Pierrot studio_Shin.Ei.Animation studio_A.1.Pictures
##          <int>          <int>          <int>
## 1:          0          0          0
## 2:          0          0          0
## 3:          0          0          0
## 4:          0          0          0
## 5:          0          0          0
## 6:          0          0          0
##  studio_DLE studio_Nippon.Animation studio_AIC studio_Tatsunoko.Production
##          <int>          <int>          <int>          <int>
## 1:          0          0          0          0
## 2:          0          0          0          0
## 3:          0          0          0          0
## 4:          0          0          0          0
## 5:          0          0          0          0

```

```

## 6:      0      0      0      0      0
##  studio_Bones studio_Xebec studio_Shift studio_T.Rex studio_Gonzo
##      <int>      <int>      <int>      <int>      <int>
## 1:      0      0      0      0      0
## 2:      0      0      0      0      0
## 3:      0      0      0      0      0
## 4:      0      0      0      0      0
## 5:      0      0      0      0      0
## 6:      0      0      0      0      0
##  studio_Kyoto.Animation studio_SILVER.LINK. studio_LIDENFILMS
##      <int>      <int>      <int>
## 1:      0      0      0
## 2:      0      0      0
## 3:      0      0      0
## 4:      0      0      0
## 5:      0      0      0
## 6:      0      0      0
##  studio_AQUA.ARIS studio_Satelight studio_X..Brains.Base.. studio_Arms
##      <int>      <int>      <int>      <int>
## 1:      0      0      0      0
## 2:      0      0      0      0
## 3:      0      0      0      0
## 4:      0      0      0      0
## 5:      0      0      0      0
## 6:      0      0      0      0
##  studio_MAPPA studio_Seven producer_NHK producer_TV.Tokyo producer_Aniplex
##      <int>      <int>      <int>      <int>      <int>
## 1:      0      0      0      0      0
## 2:      0      0      0      0      0
## 3:      0      0      0      0      0
## 4:      0      0      0      0      0
## 5:      0      0      0      0      0
## 6:      0      0      0      0      0
##  producer_Lantis producer_Movic producer_Kadokawa producer_AT.X
##      <int>      <int>      <int>      <int>
## 1:      0      0      0      0
## 2:      0      0      0      0
## 3:      0      0      0      0
## 4:      0      0      0      0
## 5:      0      0      0      0
## 6:      0      0      0      0
##  producer_Pony.Canyon producer_Bandai.Visual producer_Fuji.TV producer_Dentsu
##      <int>      <int>      <int>      <int>
## 1:      0      0      0      0
## 2:      0      0      0      0
## 3:      0      1      0      0
## 4:      0      1      0      0
## 5:      0      1      0      0
## 6:      0      0      0      0
##  producer_KlockWorx producer_Kodansha producer_Shueisha producer_Sotsu
##      <int>      <int>      <int>      <int>
## 1:      0      0      0      0
## 2:      0      0      0      0
## 3:      0      0      0      0

```

```

## 4:          0          0          0          0
## 5:          0          0          0          0
## 6:          0          0          0          0
##  producer_Mainichi.Broadcasting.System producer_Pink.Pineapple
##              <int>              <int>
## 1:          0          0
## 2:          0          0
## 3:          0          0
## 4:          0          0
## 5:          0          0
## 6:          0          0
##  producer_bilibili producer_TBS producer_Tokyo.MX producer_DAX.Production
##              <int>      <int>      <int>      <int>
## 1:          0          0          0          0
## 2:          0          0          0          0
## 3:          0          0          0          0
## 4:          0          0          0          0
## 5:          0          0          0          0
## 6:          0          0          0          0
##  producer_Tencent.Video producer_BS11 producer_Sanrio producer_TV.Asahi
##              <int>      <int>      <int>      <int>
## 1:          0          0          0          0
## 2:          0          0          0          0
## 3:          0          0          0          0
## 4:          0          0          0          0
## 5:          0          0          0          0
## 6:          0          0          0          0
##  producer_Kadokawa.Shoten producer_Nippon.Television.Network producer_Genco
##              <int>              <int>      <int>
## 1:          0          0          0
## 2:          0          0          0
## 3:          0          0          0
## 4:          0          0          0
## 5:          0          0          0
## 6:          0          0          0
##  producer_Magic.Capsule producer_Crunchyroll licenser_Funimation
##              <int>      <int>      <int>
## 1:          0          0          0
## 2:          0          0          0
## 3:          0          0          0
## 4:          0          0          1
## 5:          0          0          0
## 6:          0          0          0
##  licenser_Sentai.Filmworks licenser_Discotek.Media licenser_ADV.Films
##              <int>      <int>      <int>
## 1:          0          0          0
## 2:          0          0          0
## 3:          0          0          0
## 4:          0          0          0
## 5:          0          0          0
## 6:          0          0          0
##  licenser_Media.Blasters licenser_Aniplex.of.America licenser_VIZ.Media
##              <int>      <int>      <int>
## 1:          0          0          0

```

```

## 2:          0          0          0
## 3:          0          0          0
## 4:          0          0          0
## 5:          0          0          0
## 6:          0          0          0
##  licenser_Bandai.Entertainment licenser_Geneon.Entertainment.USA
##          <int>          <int>
## 1:          0          0
## 2:          0          0
## 3:          0          0
## 4:          1          0
## 5:          0          0
## 6:          1          0
##  licenser_Crunchyroll licenser_Nozomi.Entertainment
##          <int>          <int>
## 1:          0          0
## 2:          0          0
## 3:          0          0
## 4:          0          0
## 5:          0          0
## 6:          0          0
##  licenser_Central.Park.Media licenser_Kitty.Media licenser_GKIDS
##          <int>          <int>          <int>
## 1:          0          0          0
## 2:          0          0          0
## 3:          0          0          0
## 4:          0          0          0
## 5:          0          0          0
## 6:          0          0          0
##  licenser_Critical.Mass.Video licenser_NuTech.Digital licenser_Maiden.Japan
##          <int>          <int>          <int>
## 1:          0          0          0
## 2:          0          0          0
## 3:          0          0          0
## 4:          0          0          0
## 5:          0          0          0
## 6:          0          0          0
##  licenser_Manga.Entertainment licenser_NYAV.Post licenser_NIS.America
##          <int>          <int>          <int>
## 1:          0          0          0
## 2:          0          0          0
## 3:          0          0          0
## 4:          0          0          0
## 5:          0          0          0
## 6:          0          0          0
##  stream_Crunchyroll stream_Netflix stream_HIDIVE stream_Bahamut.Anime.Crazy
##          <int>          <int>          <int>          <int>
## 1:          0          0          0          0
## 2:          0          0          0          0
## 3:          0          0          0          0
## 4:          0          0          0          0
## 5:          0          0          0          0
## 6:          0          0          0          0
##  stream_Bilibili.Global stream_Aniplus.TV stream_Laftel stream_Ani.One.Asia

```

```

##           <int>           <int>           <int>           <int>
## 1:           0           0           0           0
## 2:           0           0           0           0
## 3:           0           0           0           0
## 4:           0           0           0           0
## 5:           0           0           0           0
## 6:           0           0           0           0
##   stream_Bilibili stream_Muse.Asia stream_Anime.Digital.Network
##           <int>           <int>           <int>
## 1:           0           0           0
## 2:           0           0           0
## 3:           0           0           0
## 4:           0           0           0
## 5:           0           0           0
## 6:           0           0           0
##   stream_CatchPlay stream_MeWatch stream_iQIYI stream_Aniplus.Asia
##           <int>           <int>           <int>           <int>
## 1:           0           0           0           0
## 2:           0           0           0           0
## 3:           0           0           0           0
## 4:           0           0           0           0
## 5:           0           0           0           0
## 6:           0           0           0           0
##   stream_Shahid stream_Sushiroll stream_Hulu stream_Animax.Korea
##           <int>           <int>           <int>           <int>
## 1:           0           0           0           0
## 2:           0           0           0           0
## 3:           0           0           0           0
## 4:           0           0           0           0
## 5:           0           0           0           0
## 6:           0           0           0           0
##   stream_Animekey
##           <int>
## 1:           0
## 2:           0
## 3:           0
## 4:           0
## 5:           0
## 6:           0

```

```

anime_targets <- details_base %>%
  select(anime_mal_id, score) %>%
  rename(
    Y_score_global = score
  )

head(anime_targets)

```

```

##   anime_mal_id Y_score_global
##           <int>           <num>
## 1:       59356             NA
## 2:       56036             6.53
## 3:        2928             6.65
## 4:        3269             7.06
## 5:        4469             6.35

```

6: 454 6.09

characters.csv

```
characters <- fread("characters.csv")
names(characters) <- str_replace_all(names(characters), "\\.", "_")

characters <- characters %>%
  mutate(character_mal_id = as.character(character_mal_id))

char_dict <- characters %>%
  transmute(
    character_mal_id,
    character_favorites = to_num(favorites),
    about_length = ifelse(is.na(about), 0, nchar(about))
  ) %>%
  left_join(char_nick_feat, by = "character_mal_id") %>%
  distinct(character_mal_id, .keep_all = TRUE)

char_works <- char_works %>%
  mutate(character_mal_id = as.character(character_mal_id))

char_to_anime <- char_works %>%
  select(anime_mal_id, character_mal_id, role) %>%
  left_join(char_dict, by = "character_mal_id") %>%
  group_by(anime_mal_id) %>%
  summarise(
    sum_character_favorites = sum(character_favorites, na.rm = TRUE),
    avg_character_favorites = mean(character_favorites, na.rm = TRUE),
    max_character_favorites = ifelse(
      all(is.na(character_favorites)),
      NA_real_,
      max(character_favorites, na.rm = TRUE)
    ),
    avg_nicknames_per_character = mean(num_nicknames, na.rm = TRUE),
    .groups = "drop"
  )
head(char_to_anime)
```

```
## # A tibble: 6 x 5
##   anime_mal_id sum_character_favorites avg_character_favorites
##         <int>             <dbl>             <dbl>
## 1           1             69689             553.
## 2           5             68925            1641.
## 3           6             22655             234.
## 4           7              489              9.23
## 5           8              35              3.5
## 6          15             5936             92.8
## # i 2 more variables: max_character_favorites <dbl>,
## #   avg_nicknames_per_character <dbl>
```

fav.csv

```

fav <- fread("fav.csv")
names(fav) <- str_replace_all(names(fav), "\\.", "_")

fav_user <- fav %>%
  group_by(username, fav_type) %>%
  summarise(n = n(), .groups = "drop") %>%
  pivot_wider(names_from = fav_type, values_from = n, values_fill = 0) %>%
  rename_with(~ paste0("nb_fav_", .x), -username)

head(fav_user)

## # A tibble: 6 x 5
##   username      nb_fav_anime nb_fav_character nb_fav_people nb_fav_company
##   <chr>          <int>          <int>          <int>          <int>
## 1 ""              4              0              0              0
## 2 "-----9-----" 10              0              0              0
## 3 "-----Ren-----" 10             10             10              0
## 4 "----Haku----"    10             10              4              0
## 5 "----phoebelyn"   5              10              5              0
## 6 "----shia"        0              5              2              0

fav_anime <- fav %>%
  filter(fav_type == "anime") %>%
  rename(anime_mal_id = id) %>%
  mutate(anime_mal_id = as.integer(anime_mal_id))

fav_anime_genres <- fav_anime %>%
  left_join(genres_mh, by = "anime_mal_id") %>%
  group_by(username) %>%
  summarise(across(starts_with("genre_"), \(x) mean(x, na.rm = TRUE)),
    .groups="drop") %>%
  rename_with(~ paste0("pref_", .x), starts_with("genre_"))
head(fav_anime_genres)

## # A tibble: 6 x 22
##   username      pref_genre_Comedy pref_genre_Fantasy pref_genre_Action
##   <chr>          <dbl>          <dbl>          <dbl>
## 1 ""              0              1              1
## 2 "-----9-----" 0.5            0.3            0.5
## 3 "-----Ren-----" 0.222          0.222          0.444
## 4 "----Haku----"    0.111          0.111          0.111
## 5 "----phoebelyn"   0.4            0.2            0.2
## 6 "----123----"     0.3            0.2            0.1
## # i 18 more variables: pref_genre_Adventure <dbl>, pref_genre_Sci.Fi <dbl>,
## #   pref_genre_Drama <dbl>, pref_genre_Romance <dbl>,
## #   pref_genre_Slice.of.Life <dbl>, pref_genre_Hentai <dbl>,
## #   pref_genre_Supernatural <dbl>, pref_genre_Avant.Garde <dbl>,
## #   pref_genre_Mystery <dbl>, pref_genre_Ecchi <dbl>, pref_genre_Sports <dbl>,
## #   pref_genre_Horror <dbl>, pref_genre_Suspense <dbl>,
## #   pref_genre_Award.Winning <dbl>, pref_genre_Gourmet <dbl>, ...

```

ratings.csv

```
ratings <- fread("ratings.csv")
names(ratings) <- str_replace_all(names(ratings), "\\.", "_")

ratings <- ratings %>%
  transmute(
    username,
    anime_id = to_num(anime_id),
    status = tolower(status),
    score = to_num(score),
    is_rewatching = to_num(is_rewatching),
    num_watched_episodes = to_num(num_watched_episodes)
  )

ratings_scored <- ratings %>%
  filter(!is.na(score) & score >= 1 & score <= 10) %>%
  mutate(
    status = ifelse(status %in% c("completed", "watching", "dropped", "on_hold", "plan_to_watch"),
                    status, "other"),
    status_completed = as.integer(status == "completed"),
    status_watching = as.integer(status == "watching"),
    status_dropped = as.integer(status == "dropped"),
    status_on_hold = as.integer(status == "on_hold"),
    status_plan_to_watch = as.integer(status == "plan_to_watch")
  ) %>%
  select(-status)

user_activity <- ratings_scored %>%
  group_by(username) %>%
  summarise(
    user_mean_score_from_ratings = mean(score, na.rm = TRUE),
    user_nb_rated = n(),
    user_nb_completed = sum(status_completed == 1, na.rm = TRUE),
    user_nb_watching = sum(status_watching == 1, na.rm = TRUE),
    user_rewatch_rate = mean(is_rewatching > 0, na.rm = TRUE),
    user_mean_watched_eps = mean(num_watched_episodes, na.rm = TRUE),
    .groups = "drop"
  )
head(user_activity)

## # A tibble: 6 x 7
##   username          user_mean_score_from_ratings user_nb_rated user_nb_completed
##   <chr>              <dbl>          <int>          <int>
## 1 ""                8              7              2
## 2 "-----788"      8.98             64             63
## 3 "-----9-----" 6.89            593            590
## 4 "----Ren-----" 8.21            143            141
## 5 "----Blank-----" 7.08            158            146
## 6 "----Haku-----" 8.31            509            432
## # i 3 more variables: user_nb_watching <int>, user_rewatch_rate <dbl>,
## #   user_mean_watched_eps <dbl>
```

Construction du tableau anime-level

```
anime_level <- details_features %>%
  left_join(char_features,      by = "anime_mal_id") %>%
  left_join(char_to_anime,      by = "anime_mal_id") %>%
  left_join(voice_features,     by = "anime_mal_id") %>%
  left_join(person_anime_features, by = "anime_mal_id") %>%
  left_join(reco_features,      by = "anime_mal_id") %>%
  left_join(anime_targets,      by = "anime_mal_id")

anime_level_train <- anime_level %>%
  filter(!is.na(Y_score_global))

fwrite(anime_level_train, "anime_level.csv")
```

Construction du tableau user-anime

```
anime_reco_num <- details_base %>%
  transmute(
    anime_mal_id = as.integer(anime_mal_id),
    episodes     = to_num(episodes),
    year         = to_num(year)
  )

anime_reco_cat <- details_base %>%
  select(anime_mal_id, type, rating, status, season, source) %>% # + source ici
  mutate(across(c(type, rating, status, season, source), as.character)) %>% # + source ici aussi
  onehot_topk("type", "type", top_k = 10) %>% # types les 10 plus fréquents
  onehot_topk("rating", "rating", top_k = NULL) %>%
  onehot_topk("status", "status", top_k = NULL) %>%
  onehot_topk("season", "season", top_k = NULL) %>%
  onehot_topk("source", "source", top_k = 15) # 15 sources les plus fréquentes

genres_mh_small <- multihot_topk(details_base, "anime_mal_id", "genres", "genre", top_k = 30)

anime_reco <- anime_reco_num %>%
  left_join(anime_reco_cat, by = "anime_mal_id") %>%
  left_join(genres_mh_small, by = "anime_mal_id") %>%
  rename(mal_id = anime_mal_id)

setDT(anime_reco)
anime_reco[, mal_id := as.integer(mal_id)]
setkey(anime_reco, mal_id)

setDT(ratings_scored)
# Échantillonnage pour limiter l'usage RAM (2 millions de lignes)
ratings_scored_sample <- ratings_scored[sample(.N, 2e6)] # 2 million

ratings_keep <- ratings_scored_sample[, .(
  username,
```

```

    mal_id = as.integer(anime_id),
    score,
    is_rewatching,
    num_watched_episodes,
    status_completed, status_watching, status_dropped,
    status_on_hold, status_plan_to_watch
  )]

user_anime_step1 <- anime_reco[ratings_keep, on = "mal_id"]

user_anime_step1 <- as_tibble(user_anime_step1)

user_anime_level <- user_anime_step1 %>%
  left_join(profiles_user, by = "username") %>%
  left_join(user_activity, by = "username") %>%
  left_join(favs_user, by = "username") %>%
  left_join(fav_anime_genres, by = "username") %>%
  rename(Y_user_score = score)

fwrite(user_anime_level, "user_anime_level.csv")

```